

Country Head-India (Commercial), Illumina.

GLOBOCAN 2024, the global cancer burden database maintained by the International Agency for Research on Cancer (IARC) under the World Health Organization, says India recorded an estimated 1.56 million new cancer cases and around 870,000 cancer-related deaths in 2024.

"This highlights the urgency for precision medicine," Dayanand says. Illumina expects its India business to grow three to five times over the next seven years. Its contributions to the Genome India project—enabling the sequencing of over 10,000 Indian genomes—are very significant for precision-medicine foundation.

Apollo Hospitals is also building institutional genomics capacity. The Apollo Genomics Institute in Karnataka brings advanced diagnostics, counselling, and prenatal screening into mainstream care.

"We are revolutionising healthcare by making advanced genetic diagnostics accessible," says Dr Manish Mattoo, CEO for Karnataka and the Central Region. The institute embeds genomics into oncology, obstetrics, paediatrics, neurology and nephrology.

Even at the clinician level, the movement is visible. Dr. Nathany notes that sequencing has become integral to patient stratification, trial recruitment and biomarker-driven therapy decisions. His laboratory operates at 70% utilisation.

Can India build affordable, equitable nationwide genomics?

By 2030, genomics is expected to be embedded across mainstream clinical care, becoming a routine component of disease detection, therapy selection, newborn screening, and preventive health. Whether India can build affordability, access, and data sov-



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CEO, STRAND LIFE SCIENCES



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SURENDRAN CHEMMENKOTIL
MD, METROPOLIS HEALTHCARE

ereignty into this vision depends on a combination of factors.

On the affordability front, the market faces persistent challenges. Tests ranging from ₹20,000 to over ₹1 lakh are still out of reach for much of India's population, and insurance coverage remains limited. Unless genomic tests enter national insurance schemes like Ayushman Bharat, the market risks entrenching an urban-elite skew.

Acharya of Mapmygenome notes that cost and interpretation remain barriers. "Clinical validation and cost are the biggest challenges to wider adoption," she says.

Regulation and data governance are equally critical. Investor caution persists around India's evolving data-privacy regime, especially with the Digital Personal Data Protection Act awaiting enforcement. Cross-border data-transfer

restrictions complicate global collaborations. And while national genomic databases are expanding, they remain early-stage.

Yet, the optimism among clinicians, investors and sequencing companies is striking.

"By 2030, genomics will be embedded across India's healthcare pathways," says Chemmenkotil of Metropolis. Illumina sees India as a rising global genomics hub.

Hariharan of Strand believes India is evolving into an innovation hub, not just a sequencing destination. Dr Nathany sees India transforming from a user of global genomic tools to a creator.

"We envision India evolving from a user of global genomic solutions to a global leader in developing them," he says **BT**

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